

WEEK 1 TASK

Developing a Data Exploration Strategy Roadmap

Python-Based Data Science Project Planning

Project Theme: Student Academic Performance Analysis

Prepared for: Week 1 Data Exploration Assignment

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Project Repository: <https://github.com/Nupur125/data-exploration-strategy>

Strategy roadmap covering objectives, hypotheses, public data sourcing, methodology, timeline, resources and risk management.

1. Introduction

Data exploration is the first practical stage of a data science project. Before building a model or drawing conclusions, the dataset needs to be understood: what each variable means, whether values are missing, whether there are unusual observations, and which patterns are worth investigating. A structured exploration plan reduces the chance of making decisions from incomplete or misleading data.

For this project, the selected problem is Student Academic Performance Analysis. The aim is to explore how study habits, attendance, previous performance, internet access and extracurricular participation relate to students' final academic performance. The project is intentionally focused first on exploration rather than prediction, so that the findings can be understood before any machine-learning stage is considered.

The complete project workflow and supporting files are maintained in the public GitHub repository: <https://github.com/Nupur125/data-exploration-strategy>.

Why this problem?

- Student performance data contains a mixture of numerical and categorical variables, making it suitable for practical exploratory data analysis.
- It allows comparison of several factors instead of focusing on a single score.
- The findings can be presented through understandable charts and summary statistics.
- The same workflow can later be adapted to other public datasets.

2. Project Objectives

The main objective is to create a clear, reproducible Python-based exploration workflow that identifies meaningful patterns in student academic performance.

Specific objectives

- Understand the structure, size and quality of the selected public dataset.
- Measure missing values, duplicates and possible data-entry inconsistencies.
- Summarize important numerical variables using descriptive statistics.
- Study the relationship between attendance, study time, previous scores and final percentage.
- Compare performance across relevant categorical groups.
- Detect possible outliers and decide whether they are genuine observations or data-quality issues.
- Create clear visualizations that communicate the strongest patterns.
- Document findings and limitations so that the analysis can be reproduced.

Key questions

- Does higher daily study time appear to be associated with better final performance?
- Is attendance related to students' final percentage?
- How strongly does previous-year performance relate to current performance?
- Do students with internet access show different performance patterns?

- Is there an observable difference between students who participate in extracurricular activities and those who do not?
- Which variables deserve further investigation in a later predictive model?

3. Hypotheses

The following hypotheses will guide exploration. They are not treated as facts; the analysis will determine whether the available data provides supporting evidence.

ID	Hypothesis	Expected observation
H1	Study hours are positively associated with final percentage.	Students with more study hours may show higher average final percentages.
H2	Attendance is positively associated with final percentage.	Higher-attendance groups may have better average performance.
H3	Previous-year score is positively associated with final percentage.	Students with stronger previous performance may tend to perform better again.
H4	Internet access may be associated with academic performance.	The two groups may show different score distributions, although the result cannot prove causation.
H5	Extracurricular participation may be associated with performance.	Performance may differ between participating and non-participating groups.

4. Data Sourcing Strategy

The dataset will be obtained from a publicly accessible source such as Kaggle, the UCI Machine Learning Repository, government open-data portals, or another reputable open-data repository. The final source will be selected using the following criteria:

- Clear description of the variables and their meanings.
- Public availability and permission to use the data for an academic project.
- Enough observations to support meaningful comparisons.
- Relevant fields such as study hours, attendance, previous score and final performance.
- Downloadable format such as CSV or Excel.
- Evidence of reasonable data quality and a traceable source.

Planned acquisition process

- Search public repositories using terms such as student performance, academic achievement and education dataset.
- Read the dataset description and data dictionary before downloading.
- Record the source name, dataset title, access date and license/usage information.
- Keep the original downloaded file unchanged and create a separate working copy.
- Load the data into Python and perform an initial validation before analysis.

5. Tools and Methodology

Tool / Library	Purpose	Reason for selection
Python	Main analysis environment	Widely used for data analysis and easy to reproduce.
Pandas	Loading, cleaning and manipulating tabular data	Provides practical DataFrame operations and grouping functions.
NumPy	Numerical operations	Useful for calculations, arrays and handling numerical values.
Matplotlib	Basic charts	Flexible plotting library for creating controlled visualizations.
Seaborn	Statistical visualizations	Makes distributions, box plots and relationship plots easier to interpret.
Jupyter Notebook	Analysis documentation	Allows code, outputs and explanations to stay together.
Optional: Scikit-learn	Later predictive modelling	Can be introduced only after exploration if prediction becomes a project goal.

6. Analysis Phases

Phase 1 — Dataset Understanding

Load the dataset, inspect rows/columns, review data types, understand the data dictionary and define the target/important variables.

Phase 2 — Data Quality Check

Check missing values, duplicate records, invalid categories, impossible values and inconsistent formatting.

Phase 3 — Data Preparation

Standardize column names, correct data types, handle missing values using a documented rule, and avoid changing genuine observations without justification.

Phase 4 — Univariate Exploration

Examine each important variable separately using counts, mean/median, standard deviation, histograms, bar charts and box plots.

Phase 5 — Bivariate Exploration

Compare final percentage with study hours, attendance and previous score; compare group-based variables using box plots and grouped summaries.

Phase 6 — Multivariate Exploration

Use correlation analysis and selected plots to see whether several variables show patterns together.

Phase 7 — Hypothesis Review

Compare the observed results with H1–H5. Distinguish association from causation and note results that need further testing.

Phase 8 — Communication

Prepare a concise findings section, charts, limitations and recommendations for a possible next stage.

Planned outputs

- A cleaned working dataset with documented changes.
- Exploratory Jupyter Notebook containing code and explanations.
- Summary statistics table.
- Charts showing distributions, group comparisons and relationships.
- Hypothesis review table.
- Short final findings and limitations section.
- All materials are version-controlled and published at <https://github.com/Nupur125/data-exploration-strategy>.

7. Roadmap and Timeline

The assignment is planned for approximately 30–35 hours. The schedule below divides the work into manageable checkpoints.

Period	Activity	Milestone / Deliverable	Est. effort
Days 1–2	Define problem, questions and hypotheses	Approved project scope + hypothesis list	4 h
Days 2–3	Search and evaluate public datasets	Selected dataset + source notes	4 h
Days 3–4	Load and inspect data	Initial profiling notebook	4 h
Days 4–5	Data quality and preparation	Clean working dataset + cleaning notes	5 h
Days 5–6	Univariate analysis	Distribution and summary charts	4 h
Days 6–7	Bivariate / multivariate analysis	Relationship charts + correlation review	5 h
Day 7	Hypothesis review and documentation	Findings, limitations and final notebook	4 h
Buffer	Fix issues, review formatting and prepare submission	Final PDF/DOC + backup files	3 h

8. Resource Allocation

- Human resource: one student/analyst responsible for dataset selection, Python analysis, visualization, documentation and quality checking.
- Computing resource: personal computer capable of running Python and Jupyter Notebook; the proposed dataset is expected to be small enough for local analysis.
- Software resource: Python environment with Pandas, NumPy, Matplotlib, Seaborn and Jupyter installed.
- Storage: separate folders for raw data, cleaned data, notebooks, figures and final report.
- Time resource: approximately 30–35 hours, including a small buffer for troubleshooting and final review.
- Version control: project history and files are tracked on GitHub at <https://github.com/Nupur125/data-exploration-strategy>.

Suggested folder structure

```
project/  
  ■■■ data_raw/      original downloaded dataset  
  ■■■ data_clean/    cleaned working dataset  
  ■■■ notebooks/     Jupyter notebooks  
  ■■■ figures/       exported charts  
  ■■■ report/        final documentation
```

9. Risk Analysis and Mitigation

Risk	Possible impact	Mitigation
Poor dataset quality	Incorrect summaries or misleading charts	Profile the data first; document cleaning rules; validate ranges and categories.
Too many missing values	Loss of useful observations	Measure missingness before choosing deletion or imputation; report the decision.
Small or biased sample	Weak generalization	State the dataset limitations and avoid claiming that results represent all students.
Confusing association with causation	Incorrect conclusions	Use careful language such as 'associated with' and avoid causal claims from simple EDA.
Outliers	Distorted averages or plots	Inspect extreme values and determine whether they are valid observations before changing them.
Tool/library errors	Delay in analysis	Keep a working environment file/requirements list and save notebook versions regularly.
Time overrun	Incomplete submission	Follow the milestone schedule and use the final buffer for troubleshooting rather than adding unnecessary analyses.

10. Checkpoints and Quality Review

- Checkpoint 1: Confirm that the problem statement, questions and hypotheses are specific and measurable.
- Checkpoint 2: Confirm that the selected dataset contains the variables needed to investigate the questions.
- Checkpoint 3: Before plotting, confirm data types, missing values, duplicates and invalid ranges.
- Checkpoint 4: Review charts for readable labels, correct units and misleading scales.
- Checkpoint 5: Re-check every conclusion against the actual output rather than assuming the hypothesis is true.
- Checkpoint 6: Run the notebook from beginning to end and verify that it works without manual fixes.

11. Expected Outcome

At the end of Week 1, the project should have a well-defined problem, a suitable public dataset, a documented exploration strategy, a Python toolset, a realistic work schedule and a risk-management plan. The immediate outcome is not a machine-learning model; it is a reliable understanding of the dataset and the relationships worth investigating further.

12. Conclusion

A planned data exploration process makes the later stages of a data science project more reliable. By starting with clear questions and hypotheses, validating the source, checking data quality and moving from simple summaries to relationships between variables, the analysis can remain focused and reproducible. This roadmap provides a practical structure for completing the project within the available time while leaving room for iteration when the data reveals unexpected patterns.

Submission note: The roadmap is designed as a planning document. Actual numerical findings will be added only after the selected public dataset has been downloaded and explored.

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